Shreyas R Shetty

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Swansea, UK - open to relocation | Available immediately | Graduate Route visa valid to May 2028 - No sponsorship required

Summary

Mechanical engineer with an MSc (Merit, Swansea University) and hands-on experience in mechanical design, CAD, and finite element analysis. Comfortable moving between the design desk and the plant floor: I have run root-cause investigations on failing shop-floor equipment as well as full design-to-manufacturing-handover cycles in SolidWorks and AutoCAD. Practised at reading technical drawings, validating designs against industry standards, and working cross-functionally with manufacturing and quality teams. Quick to learn, self-directed, and looking to build a long-term career in an industrial engineering environment.

Skills

CAD & Design: SolidWorks, AutoCAD (2D/3D), CATIA V5, Fusion 360, Siemens NX CAD, Creo, GD&T, tolerance stack-up, DFM/DFMA/DFMEA, BOM & engineering change notices

Analysis & Testing: Finite element analysis (static, modal, thermal, fatigue) in Ansys Mechanical APDL and Workbench, design validation and testing, mesh sensitivity

Manufacturing & Process: Root cause analysis (5 Whys, Ishikawa), lean manufacturing, KANBAN, just-in-time, SOP development, quality control

Standards & Compliance: ASME Y14.5M, IS standards, risk assessment, compliance reporting

Programming & Tools: Python (PyQt5, NumPy, Pandas, Matplotlib), advanced MS Excel (Macros), MS Office

Experience

Application Engineer — Mechanical Design & Simulation June 2023 – Dec 2024
ProSIM R&D Pvt. Ltd. *Bengaluru, India*

- Designed components and assemblies in SolidWorks and produced detailed manufacturing drawings with BOMs in AutoCAD, taking equipment from concept through to drawing release.
- Ran FEA (static, modal, thermal, fatigue) in Ansys to validate welded and bolted structural members against ASME and IS design standards before manufacture.
- Led design reviews with internal teams and clients and managed the resulting engineering change notices through to manufacturing handover.
- Built Python tools that replaced manual engineering calculations, cutting a multi-step calculation from around 30 minutes to a few seconds, then documented and trained colleagues on their use.

Manufacturing Engineering Intern Sept – Oct 2022
Bosch Ltd. *Bidadi, India*

- Investigated recurring tooling failures on a honing machine producing diesel fuel injector barrels using 5 Whys and Ishikawa root-cause analysis, contributing to corrective actions that improved first-pass yield and extended tool life.
- Set up a KANBAN board for production monitoring and drafted standard operating procedures for shop-floor machining.

Projects

Metro Safety Gate Valve — Team Lead BEng Final-Year Project

- Led a 5-person team through design and prototyping of a safety gate valve with Python-based rail-arrival prediction; work was published at the ICGCP conference and secured competitive State Government sponsorship. Tools used: CATIA V5, SIEMENS NX, Easy EDA, Arduino

Front Wing Aerodynamics — Formula Student member Swansea University Race Engineering Team

- Designed and validated a race car front wing in SolidWorks and Ansys, then supported prototyping and assembly with the wider team. Tools used: Fusion 360, Solidworks, Ansys Workbench, Siemens NX CAD

Education

MSc Mechanical Engineering — Merit Jan 2025 – Jan 2026
Swansea University, UK

BEng Mechanical Engineering — First Class with Distinction Aug 2019 – June 2023
Sapthagiri University, India

Certifications

- Ansys (Mechanical APDL, FEA, CFD) — Wissensquelle, 2022